PREMIER UNIVERSITY, CHATTOGRAM

Department of Computer Science and Engineering

EpiScope

A Big Data Surveillance, Forecasting and Response-Planning Platform for Seasonal, Tropical and Epidemic Diseases

Course: CSE 4345 - Big Data Analytics
Complex Engineering Problem (CEP) Assignment

Batch: 42 **Section:** C **Session:** 42nd

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1. Introduction and problem analysis

Bangladesh faces strongly seasonal dengue, diarrhoeal disease, influenza-like illness and geographically concentrated malaria. Routine facility, climate and logistics data are delayed and heterogeneous. EpiScope is an end-to-end prototype that ingests these signals, detects unusual rises, forecasts short-term demand, recommends allocations and presents privacy-safe evidence to national and district decision-makers. All encounter data used in this repository is synthetic.

2. CEP attributes and conflict resolution

P1 combines distributed data processing, epidemiological surveillance, optimisation and visual communication. P2 is managed through explicit sensitivity/specificity, privacy/granularity, equity/efficiency, freshness/cost and accuracy/interpretability trade-offs. P3 is addressed by threshold sweeps and backtesting. P4 addresses delayed reporting, duplicate IDs, orphan facilities and reporting artefacts. P5 applies ICD-10 and privacy principles. P6 maps functions to Ministry, epidemiology, district, hospital and public stakeholders. P7 is addressed through bronze-batch lineage from gold outputs.

3. Architecture and design decisions

The implementation follows a bronze-silver-gold lakehouse. Bronze records retain immutable raw schema plus batch metadata and SHA-256 checksum. Silver validates and de-identifies records, quarantining failures. Partitioned gold holds epidemiological-week aggregates. Analytics reads gold only; serving applies small-cell suppression and writes a compact JSON export to a D3 v7 dashboard. PySpark is the required full-scale engine; a portable pandas fallback exists solely to execute the local demonstration when PySpark is unavailable.

4. Data generation, quality, lineage and privacy

The deterministic generator accepts a seed and scale factor, produces 640 facilities across 64 districts, weather data, encounter data, stock snapshots and separate ground truth. It injects duplicate encounter IDs, orphan facilities, invalid ages and ICD codes. Twelve automated rules check identifiers, required fields, age, sex, ICD-10, temporal ordering, delay, facility reference, geography and outcome. The silver layer uses salted SHA-256 patient hashes and drops direct patient identifiers. Dashboard counts from 1 to 4 are suppressed.

5. Analytics methods

Detection uses a transparent EARS C2-inspired rolling baseline and adjustable sigma threshold. Forecasting produces a rolling baseline forecast with an 80% interval; a full submission should extend the gold feature contract with lagged rainfall and temperature. Forecast demand is converted into beds, IV saline and NS1 kit demand using explicit conversion factors. Allocation ranks severity-adjusted unmet need while keeping a population-equity floor. The seasonal calendar marks high quantile division-disease weeks for pre-positioning and prevention.

6. Evaluation and performance

The portable demo successfully generated 25,000 synthetic encounters at SF=.001, produced a 7.26 MB serving file, and was rerun without changing the 25,001-line bronze CSV (header plus 25,000 records). This confirms demo idempotency and serving-size compliance, but is not a substitute for the mandated SF=.25/.5/1 performance study or Spark UI evidence. Complete the following table only with reproducible measured values.

Scale factor	Ingest	Silver/quality	Gold	Analytics	Total
.001 demo	completed	completed	completed	completed	~19s
.25	[measure]	[measure]	[measure]	[measure]	[measure]
.5	[measure]	[measure]	[measure]	[measure]	[measure]
1	[measure]	[measure]	[measure]	[measure]	[measure]

7. Dashboard design and usability

The D3 v7 dashboard provides five coordinated views: selectable district alert tiles, observed/forecast time series with uncertainty, seasonal heatmap, stock-demand-allocation bars and ranked warnings. Selecting district or disease updates the views. The dashboard loads only aggregated JSON and shows its data freshness. A manual browser-helper failure prevented automated visual capture in this environment; complete a two-user usability check before submission.

8. Ethics, society and sustainability

False alerts waste scarce staff time; missed alerts can cost lives. The recommended allocation makes severity and equity visible rather than hiding political decisions in a black box. The system is designed for data minimisation: identifiers remain in protected bronze storage and public-facing small counts are suppressed. Models are advisory, with uncertainty and data-quality information visible. Local processing and compact aggregate serving reduce infrastructure costs.

9. Limitations and future work

The generated data is synthetic and does not establish operational effectiveness. The current demonstration does not replace validation on licensed routine data, local governance review, role-based access controls, a full weather-lag forecasting model, or a solver-backed multi-resource allocation experiment. These must be completed before operational use.

10. Detailed system architecture

EpiScope uses a medallion layout because the same incoming record serves different purposes at different stages. The raw intake layer must remain recoverable for audit; the analysis layer must be safe to join and aggregate; and the dashboard layer must be compact enough for ordinary browsers. Treating these purposes as separate layers avoids the common mistake of cleaning a source file in place and then losing the evidence needed to explain a later result.

Stage	Main responsibility	Output / control
Source intake	Accept CSV and JSON without changing source fields	Source name, load time, checksum and batch ID
Bronze	Keep an immutable record of each accepted batch	Raw encounter rows and manifest
Silver	Validate, quarantine, de-identify and conform	Salted patient hash; no direct identifier
Gold	Aggregate by district, disease and epi-week	Weekly case marts, quality report and features
Analytics	Create alerts, forecasts and allocation inputs	Decision-ready district measures
Serving	Apply final suppression and export compact JSON	D3 dashboard data below 10 MB

The documented DAG is generator -> pipeline -> analytics -> serving. A successful batch may be rerun because the manifest records a deterministic checksum. If the checksum already exists, bronze does not append the same file. Silver and gold are then rebuilt from immutable bronze data so a late report can change the correct historical epidemiological week.

11. Data contract and synthetic-data design

The project uses synthetic health records because the course brief prohibits releasing real line-level patient data. The generator is deterministic: the same seed and scale factor recreate the same files. This property is important for debugging, evaluation and a hidden incremental drop. At full scale, SF=1 is intended to represent at least 25 million encounters from 2021 to 2025; the verified local demonstration used SF=.001 and 25,000 rows.

Dataset	Key fields	Purpose
Facility registry	facility_id, district, division, beds, ICU beds	Reference join and capacity context
Population / district reference	district, division, population	Rate denominators and equity share
Weather	district, date, rainfall, temperature, humidity	Lagged outbreak covariates
Encounter line list	onset/report date, disease code, severity, outcome	Surveillance signal
Stock snapshot	facility, IV saline, NS1, ORS, occupied beds	Demand gap and response plan
Ground truth outbreaks	location, disease, date, duration, multiplier	Evaluation only; never used for operational alerting

The generator deliberately inserts realistic defects: approximately two percent duplicate submissions, orphan facility references, invalid ages, invalid codes, delayed reports and an artefact that can resemble an outbreak. These defects are not accidental noise. They are test cases for the quality and alerting design.

12. Data quality, privacy and lineage controls

Quality checks are applied before a record becomes available for surveillance. The project does not silently fix invalid source records; it writes failed rows to a quarantine output together with an explicit reason code. This makes data loss visible to both engineers and decision-makers.

Control family	Examples implemented	Why it matters
Schema / completeness	Encounter ID, patient ID, facility, onset date present	Prevents ambiguous and unusable records
Validity	Age 0-120, sex in F/M, approved ICD-10 code, valid outcome	Stops implausible clinical values
Temporal consistency	Onset not after report; delay no more than 21 days	Protects week-based surveillance
Reference integrity	Facility must exist in registry; district must be present	Avoids incorrect geographic attribution
Duplicate control	Latest record retained per encounter ID	Supports idempotent loads
Privacy	Salted SHA-256 hash; age retained as a non-identifying field	Keeps direct identity out of silver
Serving suppression	Counts 1-4 become Suppressed	Reduces small-cell disclosure risk

Lineage is retained by carrying bronze_batch_ids into gold aggregates. An analyst investigating an alert can therefore identify which intake batch contributed to the displayed number. In a production deployment, the salt would be injected from a secret manager rather than an environment file and access to bronze would be restricted.

13. Outbreak detection and forecasting approach

The first operational detector is deliberately interpretable. For each district and disease, EpiScope computes an eight-week rolling baseline from prior observations and compares the current count against baseline plus a configurable number of standard deviations. The dashboard default is two standard deviations. This provides an understandable starting point for discussion with surveillance officers: a warning means the recent count is materially above the district's own recent pattern, not merely above a national average.

The threshold must be selected using evidence. A low threshold identifies events earlier but increases false alerts. The evaluation plan therefore records sensitivity, false alerts per district-year and median delay to detection at several thresholds. Ground truth is isolated from the production detector and read only by the evaluation routine.

The forecast module produces one-week-ahead values and an 80 percent interval from the rolling seasonal baseline. The full Spark experiment should add weather features such as rainfall and temperature lagged four to eight weeks, then compare the model with a seasonal-naive baseline over the last 52 weeks. Evaluation must report MAE and interval coverage; accuracy alone is insufficient when officials need to plan beds and supplies.

Forecast output	Conversion assumption used in prototype	Decision use
Cases	Rolling baseline with recent signal	Expected surveillance burden
Beds	12 percent of forecast cases	Hospital surge planning
IV saline	0.7 unit per forecast case	Procurement / transfer need
NS1 tests	0.5 unit per forecast case	Dengue diagnostic readiness

14. Allocation and preparedness planning

The allocation decision is based on forecast demand, available stock, severity and an equity floor. A purely efficiency-oriented allocation may send everything to the largest city while leaving lower-risk districts without a minimum emergency reserve. EpiScope therefore exposes the policy choice instead of hiding it: severity raises priority, while population share prevents a district from being excluded completely.

Let x(d,r) be the amount of resource r allocated to district d and u(d,r) its unmet demand. The planned optimisation minimises the weighted sum of unmet demand, subject to a fixed budget, non-negative allocations, a minimum population-linked equity share and division transport capacity. The prototype produces a transparent severity-weighted allocation; the final scale run should solve and report the complete linear programme with an open-source solver.

Budget scenario	Question to answer	Required evidence
Low budget	Which minimum reserves should be protected?	Unmet demand and minimum share
Moderate budget	Where does the marginal unit avert most unmet need?	Efficiency versus equity curve
High budget	When is stock coverage sufficient across regions?	Coverage and transport constraints
Seasonal calendar	When should prevention begin?	Division-disease high-risk epi-weeks

The preparedness calendar is derived from average disease activity by division and epidemiological week. Weeks in the upper seasonal range prompt supply pre-positioning and prevention activity, while lower weeks retain routine readiness.

15. Evaluation, scalability and reproducibility

A reproducible project must tell a reviewer exactly how to rebuild the result. The repository pins its Python dependencies, exposes one orchestration command and documents the hidden-drop path. The documented run generates sources, executes the layered pipeline, calculates analytics and exports dashboard JSON. In the verified local demo, the same 25,000-row batch was run twice and bronze remained at 25,001 lines including the header. The serving export was 7.26 MB, below the stated browser limit.

Evaluation area	Metric	Status in this repository
Detection	Sensitivity, false alerts per district-year, median delay	Framework implemented; full threshold sweep pending Spark scale run
Forecast	MAE and 80 percent interval coverage versus seasonal-naive	Protocol documented; full backtest pending
Allocation	Weighted unmet demand versus population-proportional baseline	Prototype output available; solver experiment pending
Quality	Rows in, valid rows, quarantined rows, rules evaluated	Generated per load
Performance	Runtime and output size at SF=.25, .5, 1	Template prepared; must be measured on target hardware

Two Spark optimisations are specified for the final experiment: broadcast the relatively small facility registry to avoid a large shuffle join, and partition gold data by epidemiological year and division while enabling adaptive query execution for skew. Before-and-after runtimes and genuine Spark UI screenshots must be included before submission.

16. Dashboard rationale, ethics and limitations

The dashboard is designed around one decision question: where should attention and supplies go this week? A district selector and disease selector coordinate the five views. The map-like district strip supports quick comparison, the time series explains whether a warning is unusual and how uncertain the next week is, the heatmap shows seasonal recurrence, the allocation chart shows the stock gap, and the ranked table supports a short operational briefing.

Design decision	Reason
D3 v7 only	Meets the visualisation constraint and keeps interactions inspectable in source code
Aggregated JSON only	Prevents browser exposure of raw encounter data and keeps load size manageable
Uncertainty band	Avoids presenting forecasts as certain facts
Small-cell suppression message	Makes privacy transformation visible rather than silently converting small values to zero
Freshness text	Helps users judge how current the displayed data is

The system is not a clinical decision-maker. Changes in reporting practice, facility silence, testing access or coding can mimic disease change. Recommendations must be reviewed by public-health staff, and the data-quality report should be consulted before escalation. The next development stage should include role-based views, a true distributed object-store deployment, user testing with district staff and an independently reviewed privacy impact assessment.

Appendix A - Policy brief

Act earlier, allocate fairly. EpiScope combines routine reports, weather and stock data to support dengue, diarrhoeal-disease, influenza-like-illness and malaria preparedness. The top recommendations are: verify high-priority warnings quickly; pre-position supplies where forecast demand exceeds stock; and start prevention work before locally high-risk seasonal weeks. The dashboard shows observed reports, forecasts, uncertainty, stock and allocations together. Suppressed counts are a privacy safeguard, not zero cases. Forecasts remain uncertain and must support, not replace, clinical and epidemiological judgement. Direct identifiers do not leave protected ingestion storage and the dashboard receives only aggregated outputs.

Appendix B - Contribution matrix

Work area	Member 1	Member 2	Member 3
Problem analysis	[L/S/-]	[L/S/-]	[L/S/-]
Data engineering	[L/S/-]	[L/S/-]	[L/S/-]
Analytics	[L/S/-]	[L/S/-]	[L/S/-]
Dashboard	[L/S/-]	[L/S/-]	[L/S/-]
Agreed share	[%]	[%]	[%]
Signature	[sign]	[sign]	[sign]

Appendix C - Individual reflections

Member 1: [half-page reflection].

Member 2: [half-page reflection].

Member 3: [half-page reflection].